

VirSME

COMPLETE FUNCTIONAL INVENTORY

Senior product architecture, business analysis and software audit report

Audit date: 16 July 2026

17	69	137	85
Product modules	Submodules	Functions catalogued	Live functions

Implementation status distribution

Live	Partial	Prototype	Stub	Disabled	Planned	Unclear
85	35	3	4	3	2	5

Audit basis

Static, read-only inspection of all repositories under Full stack; frontend routes and API calls were traced to Express routes/controllers, Mongoose models, middleware, schedulers, Socket.IO handlers, and supporting documentation. No runtime environment, database, mail server, OpenAI account, or external messaging provider was available, so runtime-dependent items are identified for manual verification.

Source completeness notice

The supplied Markdown report contains the executive summary, master module inventory, and detailed module-by-module analysis (Sections 1–3). The requested audit brief also calls for Sections 4–10; those sections were not present in the source and have not been invented in this formatted document.

Read-only audit | No code changes performed

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Status definitions

- **Live** — a coherent UI/API/data journey exists with appropriate implementation evidence.
- **Partially implemented** — substantial implementation exists, but the journey, integration, validation, or coverage is incomplete.
- **Prototype** — demonstration/testing-oriented implementation or hard-coded sample behaviour.
- **Stub or placeholder** — referenced UI/API surface without complete business logic.
- **Disabled** — implementation exists but is explicitly unavailable or not invoked.
- **Planned** — described as future work without an implementation.
- **Unclear** — static evidence is insufficient; runtime or business verification is required.

1. Executive summary

Scope and totals

Measure	Result
Frontend applications	7 (CRM, Connect, People, Employee_dashboard, admin_dashboard_hr, task_dashboard, complete-profile)
Backend applications	1 active Express/MongoDB backend (backend); backend-hr is empty
Product modules identified	17
Submodules identified	69
Business functions catalogued	137
Live	85
Partially implemented	35
Prototype	3
Stub or placeholder	4
Disabled	3
Planned	2
Unclear	5

These counts are business capabilities, not raw source functions or HTTP endpoints. Repeated implementations in the four employee-facing portals count once unless they provide a materially different journey.

Main strengths

- 1. A broad employee lifecycle is represented end to end: candidate onboarding, employee master data, organisation structure, attendance, leave, payroll, documents, promotion, and offboarding.
- 2. The communication suite is unusually deep: internal chat, spaces, threaded discussions, supervised assignment email, WhatsApp-style conversations, attachments, reactions, read state, scheduling, and approval flows.
- 3. Employee self-service and a unified request centre cover leave, attendance challenges, salary changes, loans, advances, reimbursement, bonus, commission, documents, overtime, tax adjustments, encashment, carry forward, and whistle reports.
- 4. Payroll has substantive calculation support: periods, estimates, salary fields, tax, PF, gratuity, loans, late/half-day deductions, scheduled allowances, encrypted slips, PDFs, and email delivery.
- 5. Automation is connected to business data: session timeout/logout, absence backfill, leave-to-attendance sync, probation leave approval generation, scheduled messages, yearly leave reset, email ingestion/classification, and CNIC extraction.

Major gaps

- 6. Permission control is fragmented across fixed page-role records, Employee.permissions, isAdmin, and separate attendance/payroll/CRM grants. Several broad middleware decisions do not enforce the configured scope at route level.
- 7. The application is highly duplicated. CRM, Connect, People, and Employee Dashboard contain copies of payroll, leave, requests, messaging, chat, feedback, and task UI, creating drift and inconsistent journeys.
- 8. There is no meaningful automated test suite, OpenAPI contract, migration discipline, or CI evidence. The backend npm test script explicitly reports that no tests are specified.
- 9. Some security-sensitive design choices require urgent review: tracked .env files, globally unique employee email rather than tenant-scoped identity, permissive upload serving, JWT-in-query support, and uneven validation/error handling.

10. Several visible features are incomplete: employee PDF export, leave PDF download, ClickUp-style alternate task views, task AI description, some “My Clients” sections, and salary-structure integration into payroll.

Areas requiring manual verification

- Login, 2FA, trusted-device, refresh-token rotation, session reactivation, and forced logout using deployed credentials.
- Tenant isolation on every list/detail/update endpoint, especially delegated payroll/attendance users and employee-directory access.
- SMTP, IMAP, offer/onboarding mail delivery, scheduled mail, and sender-side popup/error behaviour.
- Real external WhatsApp provider connectivity; the code clearly implements a VirSME conversation layer, but the provider boundary is not clear.
- OpenAI model availability and CNIC/email-classification accuracy, cost, consent, and fallback behaviour.
- Payroll calculations against approved finance examples, encrypted slip recovery, PDF rendering, and end-to-end email delivery.
- Attendance time-zone, cross-midnight, browser-close, leave-sync, absent-backfill, and deduction edge cases.
- Responsive/mobile layout on all seven frontends; responsive components exist, but no device test evidence exists.

2. Master module inventory

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
Identity & access	Admin authentication	Admin login and JWT refresh	Authenticate owners/admin/HR and renew sessions	Admin, HR	Live	backend/src/routes/auth.js; backend/src/controllers/authController.js; admin App.tsx
Identity & access	Admin authentication	Password reset	Email reset token and accept a new password	Admin, HR	Live	/api/auth/forgot-password, /reset-password; auth pages in admin apps
Identity & access	Employee authentication	Employee login and confirmation code	Authenticate employee and confirm login challenge	Employee	Live	backend/src/routes/empAuth.js; portal Login.tsx, EmployeeConfirmCode.tsx
Identity & access	Employee authentication	Employee logout, heartbeat, session reactivation	Maintain and close employee work sessions	Employee	Live	/api/emp-auth/logout, /heartbeat, /reactivate-session; EmployeeSession
Identity & access	Strong authentication	Two-factor setup, confirmation and trusted devices	Add TOTP/verification and device trust	Admin, HR	Live	/api/auth/2fa/*; TwoFactorSettings.tsx; TrustedDevice
Identity & access	Roles	Page visibility/edit permission configuration	Configure hidden/view/edit by fixed role	Super-admin	Partially implemented	Pages, RolePermission; /api/role, /api/pages; enforcement is mostly frontend
Identity & access	Delegation	Attendance access grants	Delegate scoped attendance view/mark/edit/approve	Admin	Partially implemented	AttendanceAccess; attendanceAccess.js; scope enforcement is inconsistent
Identity & access	Delegation	Payroll access grants	Delegate scoped payroll view/slips/edit/lock	Admin	Partially implemented	PayrollAccess; payrollAccess.js; anyPayrollAuth.js
Identity & access	Delegation	CRM access grants	Grant all-or-nothing CRM access	Admin	Live	CRMAccess; crmAccess.js; CrmAccessGate.tsx
HR core	Employee records	Create and list employees	Maintain the workforce directory	Admin, HR	Live	Employees; /api/employees; admin Employees.tsx
HR core	Employee records	View/edit employee profile	Manage personal, contact, employment, salary and bank data	Admin, HR	Live	/api/employees/:id; EmployeeEdit.tsx; employeeSalary.js
HR core	Employee records	Bulk activate/resign	Change employment state for multiple employees	Admin, HR	Live	/api/employees/bulk/activate, /bulk/resign
HR core	Employee records	Trash, restore and permanent delete	Soft-delete, recover, or purge employee records	Admin	Partially implemented	employee routes and Employees.tsx; purge/audit controls need runtime validation
HR core	Employee lifecycle	Automatic offboarding	Mark employees offboarded based on leaving date	Admin, system	Live	/api/employees/auto-offboard; autoOffboardEmployees
HR core	Self-service profile	View own profile	Employee access to shared master record	Employee	Live	portal Profile.tsx; /api/emp-auth/me, employee endpoints
HR core	Profile revisions	Submit, edit, withdraw profile changes	Request controlled changes to master data	Employee	Live	ProfileRevision; empProfileRevisions.js; ProfileRevisions.tsx
HR core	Profile revisions	Approve/reject revisions	Apply approved changes to employee master	Admin, HR	Live	adminProfileRevisions.js; AdminProfileRevisions.tsx
HR core	Promotions	Single employee promotion	Change designation/department and compensation	Admin, HR	Live	promotionController.promoteEmployee; admin Promotion.tsx
HR core	Promotions	Bulk promotions	Apply promotion changes to a cohort	Admin, HR	Live	/api/promotion/bulk-promote; bulkPromoteEmployees
HR core	Milestones	Birthdays and work anniversaries	Surface upcoming workforce milestones	Admin, employee	Live	employee/event/upcoming routes; dashboard widgets

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
Organisation	Structure	Departments	Create, reorder, edit and remove departments	Admin, HR	Live	Department; /api/departments; Departments.tsx
Organisation	Structure	Subdepartments and designations	Maintain lower-level structure and job titles	Admin, HR	Live	SubDepartment, Designation; corresponding routes/components
Organisation	Hierarchy	Organisation chart	Define root manager and reporting tree	Admin, HR	Live	OrgHierarchy; /api/org-hierarchy; Hierarchy.tsx
Organisation	Hierarchy	Employee hierarchy	Resolve supervisors, seniors and juniors for approvals	Managers, system	Live	EmployeeHierarchy; hierarchyController; manager/leave logic
Organisation	Supervision	Assign manager/client supervision	Allocate employees and clients to managers/team leads	Admin, manager	Live	/api/manager/*; ManagerAssignments.tsx; managerController.js
Attendance	Time capture	Check in/check out	Record employee login/logout and worked time	Employee	Live	/api/emp-attendance; EmployeeSession, Attendance; portal header
Attendance	Time capture	Heartbeat/browser-close detection	Track activity and auto-close stale sessions	Employee, system	Partially implemented	heartbeat plus 10-second test cron; browser/network edge cases untested
Attendance	Records	View own attendance and statistics	Show personal records and aggregates	Employee	Live	/api/attendance/employee/:id; /stats; portal attendance utilities
Attendance	Administration	View, create, edit and delete attendance	HR management of attendance records	Admin, HR/delegate	Live	/api/admin/attendances; admin Attendance.tsx
Attendance	Configuration	Shift definitions and multi-shift setting	Configure work windows and assignments	Admin, HR	Live	Shift; /api/admin/shifts; ShiftsSettings.tsx
Attendance	Configuration	Attendance rules and flags	Configure late/absence/manual marking behaviour	Admin	Live	AttendanceConfig, AttendanceFlag; settings UI/routes
Attendance	Calendar	Non-working days	Maintain holidays/off-days used by attendance/payroll/leave	Admin	Live	payroll-period non-working-day APIs; NonWorkingDays.tsx
Attendance	Exceptions	Attendance challenge submission/withdrawal	Ask HR to correct an attendance record	Employee	Live	/api/emp-attendance/challenge; challenge forms/history
Attendance	Exceptions	Challenge review and correction	Approve/reject proposed attendance changes	Admin, HR/delegate	Partially implemented	admin attendance challenge routes/UI; corrected times require regression test
Attendance	Audit	Attendance and change logs	Review marking and modification history	Admin, HR	Live	AttendanceLog, AttendanceChangeLog; admin log/audit pages
Attendance	Audit	Between-shift logs	Record logins between configured shifts	Admin, HR	Partially implemented	between-shift route/page; rule accuracy requires runtime validation
Attendance	Automation	Daily absent backfill	Mark unrecorded employees absent unless manual mode	System	Live	index.js 23:59 cron; backfillForDate
Leave	Application	Apply for paid/unpaid/half-day leave	Submit date selection, reason and evidence	Employee, manager	Live	ApplyLeave; /api/apply-leave; portal Leave.tsx
Leave	Application	Edit/cancel leave and discuss request	Amend/withdraw and exchange messages	Employee, approver	Live	apply-leave update/cancel/messages endpoints
Leave	Approval	Hierarchical approve/reject	Route through supervisor chain to final decision	Manager, HR, admin	Live	leaveController; hierarchy refs; notifications
Leave	Balances	View and administer leave balance	Track total, paid/unpaid use and bonus	Employee, HR	Live	LeaveYearBalance, LeaveTransaction; summary APIs/UI
Leave	Policy	HR policy rule parsing	Derive notice, entitlement, sandwich, probation rules from text	System, HR	Partially implemented	utils/policyParser.js; narrow regex parser and default 22
Leave	Probation	Probation leave approval/extension	Calculate prorated entitlement and require approval	Admin, HR	Live	ProbationLeaveApproval; daily cron; approval service/UI

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
Leave	Encashment	Leave encashment request	Request cash conversion of entitlement	Employee, HR	Live	encashment model/routes; request centre
Leave	Carry forward	Carry-forward request	Request transfer of unused entitlement	Employee, HR	Live	carry-forward model/routes; request centre
Leave	Automation	Approved leave to attendance/balance sync	Update attendance/session and ledger for leave dates	System	Live	schedulers/leaveSyncCron.js daily 23:45
Leave	Automation	Bonus leave accrual	Credit days worked on non-working days	System	Disabled	schedulers/bonusLeaveCron.js exists but is not imported by active entry point
Leave	Automation	Annual entitlement reset	Reset eligible employees to configured/default entitlement	System	Live	annual index.js cron; currently hard-coded 22
Payroll	Employee salary	Salary master and custom fields	Store compensation and configurable components	Admin, HR	Live	employee-salary/salary-fields APIs; EmployeeEdit.tsx
Payroll	Structure	Salary structure configuration and preview	Define percentage/fixed components and preview breakdown	Admin	Partially implemented	SalaryStructure; config page/API; documentation says payroll integration is future
Payroll	Periods	Payroll periods and locking	Define monthly periods and processing window	Admin, payroll delegate	Live	PayrollPeriod; /api/admin/payroll-periods
Payroll	Estimates	Calculate and manage payroll estimates	Combine salary, attendance, tax, loans and adjustments	Admin, payroll delegate	Live	PayrollEstimate; PayrollEstimate.tsx; calc-preview APIs
Payroll	Estimates	Bulk estimate create/delete/send draft	Process employees as a batch	Admin, payroll delegate	Live	/api/payroll-estimates/bulk-*; employee payroll pages
Payroll	Salary slips	Generate salary slips	Persist periodic salary result	Admin, HR/payroll	Live	SalarySlip; salary-slip routes and payroll pages
Payroll	Salary slips	Generate/download PDF	Produce printable salary document	Admin, employee	Live	pdfkit, /salaryslips/pdf/:id; employee salary slip page
Payroll	Salary slips	Employee PDF export button	Export current employee payroll view	Employee	Stub or placeholder	four portal Payroll.tsx: PDF export coming soon
Payroll	Salary slips	Encrypt/decrypt salary data	Protect slip values and authorize own-slip access	Admin, employee	Partially implemented	crypto/decryption-key routes; manual key lifecycle verification required
Payroll	Salary slips	Email salary slip	Deliver PDF through SMTP	Admin, HR	Partially implemented	/api/send-slip-email; Nodemailer; external delivery unverified
Payroll	Manual payroll	Manual slip generation	Create and send an ad hoc payslip	Admin, HR	Live	admin ManualSlip.tsx; send-slip API
Payroll	Allowances/deductions	Scheduled allowances	Define recurring additions	Admin, payroll	Live	ScheduledAllowance; admin component and routes
Payroll	Allowances/deductions	Late/half-day/unpaid-leave deductions	Apply attendance-derived deductions	System, payroll	Partially implemented	lateDeductions.js, leave/session jobs; rule complexity and idempotency need tests
Payroll	Access	Employee salary-slip self-service	View own historical slips after authorization	Employee	Live	payroll-access own-slip routes; portal Payroll.tsx/SalarySlips.tsx
Finance & benefits	Loans	Loan request and approval	Submit, review and decide employee loan	Employee, HR/admin	Live	LoanRequest; /api/loan-requests; request centre
Finance & benefits	Loans	Loan setup, EMI and benefits	Configure active loan and payroll deductions	Admin, payroll	Live	LoanDetail; /api/loans; loanPayroll.js, loanDeductions.js
Finance & benefits	Requests	Advance salary	Submit and approve advance	Employee, HR/admin	Live	AdvanceSalaryRequest; routes and request centre
Finance & benefits	Requests	Reimbursement	Submit and approve reimbursable expense	Employee, HR/admin	Live	ReimbursementRequest; routes and request centre
Finance & benefits	Requests	Bonus	Submit and approve bonus request	Employee, HR/admin	Live	BonusRequest; routes and modal/request centre

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
Finance & benefits	Requests	Commission	Submit and approve commission request	Employee, HR/admin	Live	CommissionRequest; routes and CRM page
Finance & benefits	Tax	Tax brackets and automatic tax	Configure tax and calculate deductions	Admin, payroll	Live	tax routes/models; TaxManagement.tsx; auto-tax status
Finance & benefits	Tax	Tax adjustment request	Request a manual tax correction	Employee, HR/admin	Live	TaxAdjustmentRequest; request centre
Finance & benefits	Benefits	Provident fund	Configure employee/company PF and overrides	Admin, payroll	Live	ProvidentFund; PF routes and calculator
Finance & benefits	Benefits	Gratuity calculation	Calculate and persist gratuity	Admin, payroll	Live	Gratuity; routes and calculator
Recruitment & onboarding	Offers	Create/bulk-generate offer letters	Generate candidate offers from HR data	Admin, HR	Live	OfferLetter; offer-letter routes; offer pages
Recruitment & onboarding	Offers	Email offer and candidate response	Deliver offer and process reply	Admin, candidate	Partially implemented	offer SMTP + IMAP watcher; deliverability check disabled and external mail unverified
Recruitment & onboarding	Candidate intake	Send secure CNIC/CV request	Ask candidate for missing documents	Admin, HR	Live	/api/onboarding/request-cnic-cv; request pages
Recruitment & onboarding	Candidate intake	Candidate completes profile	Fill personal/employment/bank details via tokenized route	Candidate	Live	complete-profile/id; /api/employee/id/complete
Recruitment & onboarding	Candidate intake	Upload CNIC, CV, photo and certificates	Attach onboarding evidence	Candidate, HR	Live	employee-doc/certificate upload routes; complete-profile app
Recruitment & onboarding	Employee creation	Create staff/onboarding record	Convert candidate inputs into workforce record	Admin, HR	Live	/api/staff/create; OnboardingForm.tsx; Employees
Documents	Generation	HR letters and certificates	Generate experience, salary, warning and other PDFs	Admin, HR	Live	/api/generate/*; generate-pdfs.js; docs.js
Documents	Templates	Document templates and saved documents	Create/reuse HTML/PDF document definitions	Admin, HR	Live	DocTemplate; /api/docs/templates, saved-doc routes
Documents	Employee files	Store and retrieve employee documents	Manage CNIC/CV/certificates/contracts and bulk docs	Employee, HR	Live	EmployeeDoc, EmployeeDocument, Certificate; docs pages
Documents	Signatures	Company and email signatures	Configure signatures used in generated/sent material	Admin, employee	Live	signature/email-signature routes and editors
Documents	Leave document	Download leave application PDF	Give employee a PDF after leave submission	Employee	Stub or placeholder	employee/admin success UIs label download as "Coming Soon"
HR operations	Policy	HR policies	Create, edit and publish policy content	Admin, HR; employees read	Live	HRPolicy; /api/hr-policies; policy editor/view pages
HR operations	Rules	Operational HR rules	Maintain rules consumed by portal/leave	Admin, HR; employees read	Live	HRRule; /api/hr-rules; admin and portal pages
HR operations	Discipline	Warnings	Issue, view, update and acknowledge warnings	HR/admin, employee	Live	Warning; warning routes/pages
HR operations	Discipline	Penalties	Issue and manage penalties with delegated rights	HR/admin/delegate, employee	Live	Penalty; penalty routes/pages; Employee permission flags
HR operations	Probation/notice	Probation and notice settings	Configure employment thresholds	Admin, HR	Live	ProbationPeriod, notice routes/components
HR operations	Feedback	Submit bug/feedback	Capture employee issue with attachment/priority	All users	Live	Bug; /api/bugs/create; FeedbackWidget.tsx in apps
HR operations	Feedback	Triage, approve, reward and resolve feedback	Administer reported issues and balances	Admin	Partially implemented	bug routes and RequestCenterBugs.tsx; reward business rules unclear
CMS	Internal chat	Direct conversations	Start and exchange employee messages	All employees	Live	Conversation, Message; /api/chat/conversations; Google Chat UI

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
CMS	Internal chat	Spaces and membership	Create spaces, manage member roles/details	Employee, space manager	Live	Space; chat space routes; membership components
CMS	Messaging	Rich message operations	Edit/delete, react, reply, forward, star, pin, search, mention	Employees	Live	extensive chat.js routes/controller and chat components
CMS	Messaging	Attachments and shared content	Upload and browse files/images/links	Employees	Live	chat upload/shared-content endpoints and components
CMS	Messaging	Read state, unread counts and typing	Real-time awareness and notification state	Employees	Live	chat routes plus Socket.IO rooms/events
CMS	Thread chat	Contextual thread discussions	Comment around a business thread with participants	Employees	Live	ThreadChatMessage; /api/chat-threads; Socket.IO handlers
CMS	Thread chat	Thread summary	Generate/retrieve a thread summary	Employees	Partially implemented	socket get_thread_summary; algorithm/provider not clearly isolated
CMS	Email	Assignment email inbox	Inbox/sent/drafts/star/spam/trash/search/activity	CRM users, managers	Live	AssignmentMessage; 62 assignment-message routes; Email UI
CMS	Email	Compose/reply/attachments/schedule	Create, send, draft and schedule messages	CRM users	Live	assignment-message controllers/routes; scheduled-message cron
CMS	Email	Supervisory approval	Approve/disapprove/edit/resubmit before delivery	Managers/team leads	Live	pending-approval endpoints and Socket.IO approval chain
CMS	Email	External IMAP ingestion and automated reply	Receive/classify mail and update candidate/client flows	System	Partially implemented	watcher.js, multiple IMAP services; deployment and duplicate-service behaviour unverified
CMS	WhatsApp-style messaging	Direct/group conversations	VirSME-hosted client/team chat with message state	CRM users, clients	Live	WhatsApp models/controllers/routes and UI
CMS	WhatsApp-style messaging	Scheduling, approval, reactions and comments	Govern outbound messages and collaborate	CRM users, approvers	Live	47 routes, Socket.IO, scheduled cron
CMS	WhatsApp integration	External provider delivery	Send/receive through actual WhatsApp network	CRM users, clients	Unclear	no clearly identified Meta/Twilio/360Dialog provider boundary
CRM	Client records	Client CRUD and photos	Maintain customer/company contact records	CRM users, managers	Live	ClientInfo; 23 client-info routes; clients/manager pages
CRM	Client records	Company contacts	Maintain contacts within a client company	CRM users	Live	company-employee endpoints and UI
CRM	Assignment	Assign client and employees to manager	Establish service ownership/supervision	Admin, manager	Live	manager routes/controllers; assignment UI
CRM	Assignment	Assignment activity and unread state	Track new clients, reads and changes	CRM users	Live	client read/new/flags routes and Socket.IO
CRM	Communication	Client email/WhatsApp workspace	Open customer-linked communication tools	CRM users	Live	CRM Email.tsx, Whatsapp.tsx, SendQuery.tsx
CRM	Client workspace	Extended My Clients panels	Present additional client operations	Employee	Stub or placeholder	portal MyClients.tsx contains "coming soon" panels
CRM	Conversion	Convert client/opportunity to project/task	Create delivery work from CRM state	CRM/project users	Planned	no traced endpoint/model connects CRM clients to task/workspace creation
TMS	Admin workspace	Workspace CRUD and membership	Administer task workspaces and users	Admin	Live	/api/admin/workspace; admin/task TaskManagement.tsx
TMS	Admin workspace	Space CRUD and visibility	Organise task areas and visibility	Admin	Live	/api/admin/space, /visibility
TMS	Employee work	Employee workspace/space/list tasks	Create, view, edit, assign and delete tasks	Employee, manager	Live	employee task routes; duplicated WorkspaceContext.tsx
TMS	Chat-linked tasks	Tasks inside communication spaces	Use chat spaces as the task dashboard data source	Space members	Live	ChatTask; /api/chat/spaces/:id/tasks; task

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
						dashboard context
TMS	Chat-linked tasks	Multi-assignee and creator review	Assign many employees and route completion for review	Creator, assignee	Live	ChatTask.assignees/inReview; task dashboard mapping
TMS	Task detail	Attachments and comments	Collaborate on work items	Task members	Live	task attachment routes; ChatTaskComment endpoints/components
TMS	Views	List task view	Browse and update tasks in a list	Employees	Live	task list components and context
TMS	Views	Board/calendar/Gantt/other views	Alternate visualisations	Employees	Stub or placeholder	ClickupSpace.tsx: non-list views "coming soon"
TMS	Content	Folders, docs and bookmarks	Attach supporting structure to old task workspaces	Employees	Partially implemented	employee-task APIs/context; new chat-space source does not use them
TMS	AI assist	Generate task description	Draft description with AI	Employees	Planned	CreateTaskDialog.tsx TODO; no API/service traced
TMS	Activity	Communication activity feed	Show recent assignment-message activity	Employees	Partially implemented	ActivityView.tsx uses assignment activity rather than task audit
Requests & approvals	Unified inbox	Open requests	Aggregate employee requests and pending approvals	Employee, manager	Live	/api/open-requests; OpenRequestsPanel.tsx
Requests & approvals	Unified inbox	Admin request centre	Review many request types in one UI	Admin, HR	Live	RequestCenter.tsx, AllRequests.tsx, notification widgets
Requests & approvals	Workflow	Edit/withdraw generic request	Amend or withdraw supported requests	Employee	Partially implemented	generic open-request endpoints do not cover every request model uniformly
Requests & approvals	Workflow	Admin self-request auto-approval	Auto-approve/resolve requests from admins	Admin	Live	utils/requestAutoApproval.js used by request controllers
Requests & approvals	Confidential reporting	Whistle report submission/review	Submit confidential concern and process it	Employee, admin	Partially implemented	WhistleReport; routes/request UI; anonymity and confidentiality unverified
Shared platform	Tenant/company	Company profile	Store organisation branding/contact settings	Admin	Live	CompanyProfile; /api/company-profile; admin component
Shared platform	User administration	Create/manage admin users	Maintain non-employee administrative accounts	Super-admin/admin	Live	Users; role create-user endpoints; user management pages
Shared platform	Settings	Time zone, date, font and feature settings	Configure shared presentation/processing options	Admin, user	Partially implemented	settings/date/font/feature models and pages; some jobs hard-code Karachi
Shared platform	Notifications	Real-time event notifications	Notify owners, employees, managers and rooms	All users	Live	Socket.IO in index.js; widgets/unread endpoints
Shared platform	Notifications	Persistent notification centre	Retain and manage a canonical notification history	All users	Unclear	no dedicated notification model; UI derives state from domain records/socket events
Shared platform	Search	Cross-domain search	Search clients, messages, contacts and employees	Employees	Partially implemented	domain-specific search endpoints; no single global search index/UI
Shared platform	Files	Upload/download files	Store documents and attachments	All authorised users	Live	Multer middleware; /uploads; domain attachment routes
Shared platform	Import/export	Bulk document/offer processing	Process batches of employee documents/offers	HR/admin	Live	bulk docs and bulk offer endpoints/UI
Shared platform	Import/export	General employee/data CSV import/export	Move master data in/out in a supported format	Admin	Unclear	MIME support exists, but no complete general import/export journey was traced
Shared platform	Audit	Cross-system audit log	Canonical record of security and business changes	Auditor/admin	Unclear	domain logs exist; no unified immutable audit model/view
Reporting & analytics	Dashboards	Admin workforce dashboard	Show employee, attendance, leave and request metrics	Admin, HR	Live	admin Dashboard.tsx; birthdays/events/request APIs

Module	Submodule	Function	Purpose	Primary users	Status	Evidence
Reporting & analytics	Attendance	Attendance summaries and stats	Analyse employee/time-period attendance	HR, managers, employee	Live	attendance stats/routes/pages
Reporting & analytics	Leave	Leave summaries/history	Analyse balances and consumption	HR, employee	Live	leave summary/history APIs and tables
Reporting & analytics	Payroll	Payroll summaries/estimates	Analyse expected and generated payroll	Admin, payroll	Live	payroll estimate and period pages/APIs
Reporting & analytics	Export	Report exports	Export general analytical results	Admin	Partially implemented	xlsx/csv libraries and selected flows; no consistent report catalogue
AI & automation	Identity extraction	CNIC vision extraction	Parse Pakistani CNIC fields from an image	HR/system	Partially implemented	deepseekService.extractCNICUsingOpenAI; watcher.js; external model/manual validation needed
AI & automation	Email intelligence	Incoming email intent classification	Classify offer/approval/leave/HR replies	System	Partially implemented	classifyEmailUsingOpenAI; watcher.js; fallback defaults to hr_related
AI & automation	Mail response	Draft/automated replies	Produce or send contextual replies	HR/system	Prototype	services/draftReply.js, watcher response branches; production governance unclear
AI & automation	Policy intelligence	Natural-language policy understanding	Convert policy documents to executable rules	HR/system	Prototype	natural tokenizer plus narrow regex parser, not a general rules engine
AI & automation	Workflow engine	Configurable no-code automation	Let admins define triggers/actions across modules	Admin	Disabled	no general workflow-rule model/editor/executor; only hard-coded jobs/controllers

3. Detailed module-by-module analysis

To keep the evidence auditable, this section groups closely related actions into the business functions counted above. Exact routes are representative; route files contain variants for list/detail/update/delete.

3.1 Identity and access management

Module purpose

Authenticates two overlapping identity types (User and Employee), manages sessions and 2FA, controls page visibility, and delegates module access.

Users and roles

Super-admin, admin/owner, HR, manager/team lead, employee, and delegated employee. `auth.js` normalises root admins and employee-admins; `empAuth.js` handles employee sessions.

Submodules and functions

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Admin authentication/reset	Login, refresh, logout and recover password	Credentials → token → protected admin app	User role; token version	User, reset token fields	<code>/api/auth/*</code>	SMTP reset email	Live	<code>routes/auth.js</code> , <code>controllers/authController.js</code>
Employee authentication/session	Login/confirm, heartbeat, logout/reactivate	Login → confirmation → active work session	Active employee/status checks	Employee, EmployeeSession, TrustedDevice	<code>/api/emp-auth/*</code>	session timeout job	Live	<code>routes/empAuth.js</code> , portal auth pages
2FA and trust	Configure/verify/disable 2FA and trust device	Setup → confirm → session verification	Admin/User journey	User, TrustedDevice	<code>/api/auth/2fa/*</code>	email/TOTP dependency	Live	<code>TwoFactorSettings.tsx</code> , auth routes
Page RBAC	Configure hidden/view/edit per page/role	Super-admin changes page matrix; sidebars query it	Fixed four-role enum	Page, RolePermission	<code>/api/role</code> , <code>/api/pages</code>	Frontend gates	Partially implemented	<code>models/Pages.js</code> , <code>routes/role.js</code> , <code>usePagePermission.tsx</code>
Delegated access	Grant attendance, payroll or CRM rights	Admin selects employee/scope/rights → grantee accesses module	Admin grants; delegated employee consumes	AttendanceAccess, PayrollAccess, CRMAccess	respective <code>/api/*-access</code> routes	middleware checks	Partially implemented	<code>anyPayrollAuth.js</code> , <code>CombinedAccessManager.tsx</code>

3.2 HR core

Module purpose

Maintains the authoritative workforce record and lifecycle changes.

Users and roles

Admin/HR manage all records; employees view their own and request revisions; managers receive scoped roster views.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Employee CRUD/lifecycle	Create, list, edit, trash/restore, activate/resign/offboard	Admin creates/edits → employee becomes active → later offboarded	Broad authenticated access; delete should be admin-only	Employee	/api/employees/*	auto-offboard action	Partially implemented	routes/employees.js, Employees.tsx, EmployeeEdit.tsx
Employee self-profile	Read shared personal/employment information	Employee logs in → opens Profile	Own record	Employee	/api/emp-auth/me and employee endpoints	Used by all apps	Live	four portal Profile.tsx files
Controlled revisions	Submit/withdraw and approve/reject proposed changes	Employee submits diff → admin reviews → master updated	Own submission; admin review	ProfileRevision, Employee	employee/admin profile-revision routes	socket/email notices	Live	revision controllers and both UIs
Promotion	Single/bulk organisational and salary change	Select employees → specify new terms → commit	Admin/HR	Employee, salary details	/api/promotion/*	salary breakdown preview	Live	promotionController.js, Promotion.tsx
Milestones	Surface birthdays/anniversaries	Dashboard loads upcoming dates	Tenant scope	Employee, Event	employee/upcoming/event routes	notifications/widgets	Live	employee controller and dashboard

3.3 Organisation management

Module purpose

Defines organisational units, reporting relationships, supervision and approval chains.

Users and roles

Admin and HR configure; managers/team leads use supervision; all employees may view hierarchy/rosters where exposed.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Department structure	CRUD/reorder departments, subdepartments and designations	Admin settings → save structure → use on employees/offers	Admin/HR	Department, SubDepartment, Designation	/api/departments, /subdepartments, /designations	employee/offer selectors	Live	admin components and route files
Org chart	Persist root manager and reporting tree	Admin builds chart → saves bulk tree	Admin/HR	OrgHierarchy	/api/org-hierarchy/bulkCreate	approval/roster consumers	Live	Hierarchy.tsx, org hierarchy controller
Employee hierarchy	Resolve seniors/juniors/manager chain	System traverses relationships	Manager scoping	EmployeeHierarchy, Employee supervision	hierarchy/manager APIs	leave/message approval	Live	hierarchy and manager controllers
Client/employee supervision	Assign managers/team leads	Admin/manager selects assignees → saves relationship	Hierarchy-derived rights	Employee, ClientInfo	/api/manager/*	sockets notify assignment	Live	ManagerAssignments.tsx, manager controller

3.4 Attendance management

Module purpose

Captures work sessions, calculates status/time, administers corrections and feeds leave/payroll.

Users and roles

Employees capture/view own time; admin/HR and delegated attendance users administer; supervisors receive selected notifications.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Time capture/session	Check-in/out, heartbeat and reactivate	Employee checks in → heartbeat → checkout/timeout	Own employee token	Attendance, EmployeeSession	/api/emp-attendance/*, /emp-auth/heartbeat	timeout cron	Partially implemented	empAttendance.js, sessionTimeoutCron.js
Personal records/stats	View attendance history and totals	Employee opens payroll/profile attendance	Own record	Attendance	/api/attendance/employee/:id/[stats]	payroll view	Live	attendanceUtils.ts copies
Admin records	Create/edit/delete/list attendance	HR filters period → corrects record	attendance edit grant/admin	Attendance, logs	/api/admin/attendances	logging/socket	Live	admin Attendance.tsx
Shifts/config/flags	Define shifts and attendance rules	Admin configures work windows and policies	Admin/edit grant	Shift, AttendanceConfig, AttendanceFlag	admin settings APIs	time/deduction logic	Live	settings components/routes
Holidays	Manage non-working dates	Admin selects period/date → save	Admin/payroll edit	PayrollPeriod non-working dates	/api/admin/payroll-periods/non-working-days	leave/bonus/payroll	Live	NonWorkingDays.tsx
Challenge workflow	Submit/withdraw and review a correction	Employee challenges → HR approves/rejects → record updated	Own submit; approver	attendance challenge/log records	/api/emp-attendance/challenge/*, admin review	Socket notices	Partially implemented	challenge components, attendance controller
Audit/between-shift	View change and between-shift histories	Admin opens logs/audit	Admin/HR	AttendanceLog, AttendanceChangeLog	attendance log routes	generated by mutations	Partially implemented	AttendanceLogs.tsx, AttendanceAuditLog.tsx, BetweenShiftLogs.tsx
Absent backfill	Create absences for missing daily records	Nightly job checks each tenant/config	System	Employee, Attendance, config	backfillForDate	23:59 Karachi cron	Live	backend/src/index.js cron

3.5 Leave management

Module purpose

Handles employee leave requests, hierarchical approval, entitlement ledgers, probation, encashment/carry-forward, and attendance synchronisation.

Users and roles

Employees and managers apply; supervisors/HR/admin approve according to hierarchy; payroll consumes unpaid leave.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Apply/edit/cancel/chat	Submit date/type/reason, amend or discuss	Wizard → submit → track/cancel/message	Employee; own request	ApplyLeave, leave messages	/api/apply-leave/*	notifications	Live	portal Leave components, leaveController.js
Hierarchical decision	Route approvals through seniors	Next supervisor reviews → final approval/rejection	Hierarchy chain/admin	ApplyLeave, EmployeeHierarchy	approve/reject endpoints	sockets; balance deduction	Live	leave controller approval logic
Balance/ledger	Maintain total, bonus, paid/unpaid usage	Employee/HR views balance/history	Own vs admin	LeaveYearBalance, LeaveTransaction, employee entitlement	summary/history APIs	leave/payroll jobs	Live	models and leave summary routes
Policy evaluation	Enforce notice/sandwich/probation rules	Policy text parsed during validation	HR defines; system evaluates	HRPolicy, rules	policyParser.js	regex/natural	Partially implemented	parser has narrow patterns/defaults
Probation entitlement	Generate pending prorated credit, approve/reject/extend	Daily job creates approval → HR acts	Admin/HR	ProbationPeriod, ProbationLeaveApproval	probation service/routes	daily cron	Live	index.js, probation UI/service
Encash/carry forward	Submit and process leave conversions	Employee applies → request centre decides	Employee/admin	request-specific models	encash/carry-forward routes	unified inbox	Live	routes/services and AllRequests.tsx
Sync/reset/bonus	Sync approved leave, reset yearly entitlement, accrue bonus	Scheduled job updates ledgers/attendance	System	leave + attendance/session	scheduler files/index cron	jobs	Partially implemented	sync/reset active; bonus scheduler not imported; 22 hard-coded

3.6 Payroll management

Module purpose

Configures compensation, estimates payroll, applies adjustments and produces protected salary slips.

Users and roles

Admin/HR/payroll delegates process; employees see their own compensation/slips.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Salary master/fields	Store salary and dynamic components	HR edits employee compensation	payroll edit/admin	Employee, salary field models	employee-salary/salary-fields	promotion/offers	Live	salary routes, EmployeeEdit.tsx
Structure config/preview	Define fixed/percentage breakdown	Admin saves default → previews employee	Admin	SalaryStructure	/api/salary-structure/*	not yet main payroll source	Partially implemented	summary document and controller/UI
Periods	Define/lock monthly processing	Admin creates period/non-working days	payroll edit/lock	PayrollPeriod	/api/admin/payroll-periods	attendance/payroll	Live	period model/component
Estimate/bulk processing	Calculate/upsert/bulk send/delete estimates	Select period/employees → calculate → review	payroll access	PayrollEstimate, overrides	/api/payroll-estimates/*	tax/loans/attendance/leave	Live	PayrollEstimate.tsx, controller
Slips/PDF/manual/email	Generate slips and PDFs, manual slips, send mail	Finalise → generate → email/download	admin/payroll; own-slip view	SalarySlip, encryption keys	salary-slip and send-slip routes	PDFKit/Puppeteer/Nodemailer	Partially implemented	routes/pages; external email and recovery untested
Allowance/deduction	Apply recurring allowance and attendance/loan/tax deductions	Configure → estimate calculation applies	payroll admin/system	allowances, loan, attendance, tax	utilities/controllers	automatic calculation	Partially implemented	scheduled allowances, deduction utilities
Employee self-service	View/decrypt own slips	Employee selects month and authorises	Own record only	SalarySlip	payroll-access own-slip endpoints	PDF view	Live	portal Payroll/SalarySlips pages

3.7 Finance and benefits

Module purpose

Manages employee finance requests, loans, statutory/benefit calculations and tax.

Users and roles

Employees/managers submit; HR/admin decide; payroll administers deductions and benefit settings.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Loan request/setup/EMI	Approve request then configure repayment	Apply → approve → create loan → payroll deducts	employee/admin/payroll	LoanRequest, LoanDetail	loan-request/loan APIs	payroll deduction	Live	loan controllers/services/calculators
Advance/reimbursement/bonus/commission	Submit and decide common finance requests	Form → request inbox → status update	employee/admin	four request models	respective request routes	sockets/header badges	Live	request pages/controllers
Tax and adjustment	Configure automatic tax and handle correction requests	Admin defines tax; employee may request adjustment	payroll/employee	tax models, TaxAdjustmentRequest	/api/tax, adjustment routes	payroll estimate	Live	TaxManagement and request centre
PF	Calculate/configure employee/company contribution	Admin selects employee/all → saves override	payroll admin	ProvidentFund	/api/pt/*	payroll input	Live	PFCalculator.tsx
Gratuity	Calculate/persist service benefit	Admin chooses employee and terms	payroll admin	Gratuity	/api/gratuity/*	employee tenure	Live	GratuityCalculator.tsx

3.8 Recruitment and onboarding

Module purpose

Moves a candidate from offer through secure information/document collection into the employee master.

Users and roles

Admin/HR initiate; candidates use tokenised complete-profile journey; system watches replies.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Offer generation/email	Generate one/bulk offer and send	HR chooses candidate/template → PDF → email	Admin/HR	OfferLetter, templates	offer-letter and offer-email APIs	PDF + SMTP + IMAP	Partially implemented	offer pages/controllers; delivery external
Secure data request	Request CNIC/CV/profile completion	HR emails link → candidate opens token route	Tokenised candidate access	Employee, onboarding docs	/api/onboarding/request-cnic-cv, /employee/:id/complete	mail service	Live	request pages, complete-profile app
Document collection	Upload/remove CNIC, CV, photo, certificates	Candidate completes form and uploads	Candidate link/admin	employee docs/certificates	upload APIs	CNIC AI may enrich	Live	complete-profile page and upload routes
Staff creation	Create final workforce record	HR reviews/creates staff → employee record available	Admin/HR	Employee	/api/staff/create	shared with all modules	Live	OnboardingForm.tsx, staff route
Reply processing	Detect offer acceptance/rejection	Watch inbox → classify → update/respond	System	offer/employee/mail records	watcher.js	OpenAI/IMAP/SMTP	Partially implemented	watcher and deepseek service

3.9 Document management

Module purpose

Stores evidence, templates, signatures and generated HR/payroll documents.

Users and roles

HR/admin manage and generate; candidates/employees upload/view their authorised documents.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
HR document generation	Produce letters/contracts/certificates	Select employee/template → preview/generate/download	Admin/HR	templates/saved docs/employee	/api/docs/*, /api/generate/*	PDFKit/Puppeteer/pdf-lib	Live	docs.js, generate-pdfs.js
Templates/saved docs	Maintain reusable content	Admin edits template → generate document	Admin/HR	DocTemplate, saved docs	docs template routes	signatures/company profile	Live	admin Documents/HRDocument pages
Employee files/bulk docs	Upload/retrieve/delete and bulk generate	Candidate/HR uploads; employee/admin views	Own/admin	EmployeeDoc, EmployeeDocument, Certificate	employee-doc/certificate/docs bulk routes	onboarding	Live	complete-profile/admin employee pages
Signatures	Configure company and email signature	User edits → documents/messages include signature	Admin or own email signature	signature models	signature endpoints	PDF/email	Live	signature editors/dialogs
Leave PDF	Download submission as PDF	Success screen → download	Employee	none traced	no completed endpoint connection	none	Stub or placeholder	"Coming Soon" in leave success components

3.10 HR operations

Module purpose

Publishes policy/rules, manages discipline and probation/notice settings, and captures product feedback.

Users and roles

Admin/HR author; employees read or acknowledge; selected employees can receive warning/penalty administration rights.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Policies/rules	Author and present HR guidance	HR edits → employee portal displays	HR edit; employee read	HRPolicy, HRRule	policy/rule routes	leave parser	Live	policy editor and portal HrPolicy pages
Warnings/penalties	Issue/manage discipline	HR/delegate selects employee → records action	admin/permission flags	Warning, Penalty	warning/penalty routes	employee widgets	Live	pages/controllers and employee permissions
Probation/notice	Configure employment policies	HR saves durations/options	Admin/HR	ProbationPeriod, notice setting	settings routes	probation cron/leave	Live	settings components
Product feedback	Submit, prioritise, approve/reward/resolve bugs	User widget → admin feedback queue → resolution	all submit; admin manages	Bug	/api/bugs/*	balance/reward logic	Partially implemented	FeedbackWidget copies, RequestCenterBugs

3.11 Communication Management System (CMS)

Module purpose

Provides internal chat/spaces, threaded discussion, supervised email-like assignment communication, and WhatsApp-style client/team conversations.

Users and roles

Employees communicate; space managers administer members; managers/team leads approve supervised messages; CRM users communicate with assigned clients.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Direct chat/spaces	Create DMs/spaces and manage membership	Start chat/space → add members → message	member/space roles	Conversation, Space, Message	/api/chat/*	Socket.IO	Live	chat.js, chatController.js, GoogleChat components
Rich messaging/shared content	Reply/edit/delete/react/forward/star/pin/search/mention/upload	User acts on message/thread	participant/member	message/attachment records	chat endpoints	socket read/typing events	Live	~75 chat routes and UI components
Thread discussions	Context, participants, message CRUD/reactions/search/summary	Open thread → discuss/search/summarise	thread participants	ThreadChatMessage	/api/chat-threads; socket handlers	realtime	Partially implemented	duplicate join_thread_chat; summary implementation unclear
Assignment email	Inbox/drafts/sent/spam/trash, compose/schedule/attachments	Compose → supervisor approval if required → delivery	sender/recipient/chain	AssignmentMessage, labels	/api/assignment-messages/*	cron + sockets + SMTP	Live	assignment controllers, Email UI
External email ingestion	Watch mailbox and classify/process replies	IMAP sees mail → stores/classifies → notifies/replies	system credentials	assignment/offer/email records	watcher and mail services	IMAP/OpenAI/SMTP	Partially implemented	multiple overlapping receiver services; runtime dependency
WhatsApp-style chat	Direct/group CRUD, messages, seen state, reactions/comments	Open client/group → send/schedule → collaborate	participant/approver	WhatsApp message/group models	/api/whatsapp-messages/*	cron + Socket.IO	Live	controllers/routes and Whatsapp pages
Actual WhatsApp network	Deliver through WhatsApp provider	VirSME outbound → provider → recipient	CRM access	provider record unknown	provider adapter not found	external provider	Unclear	codebase names functionality but does not identify provider API

3.12 Customer Relationship Management (CRM)

Module purpose

Maintains client/contact ownership and connects each client to supervised communication.

Users and roles

CRM-granted employees, managers/team leads and admins.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Client/company contacts	CRUD client, photo and company employees	CRM user creates/updates client/contact	CRM grant/hierarchy	ClientInfo	/api/client-info/*	communication recipients	Live	MyClients/ManagerOnly/ClientsPage
Assignment/supervision	Allocate clients and team employees	Manager assigns → ownership appears in roster	hierarchy/CRM grant	ClientInfo, Employee	/api/manager/*	socket assignment updates	Live	manager controller and assignment UI
Activity/read flags	Track new/unread and WhatsApp flags	User opens client → system marks state	assigned user	client/read state	client-info flag/read endpoints	sockets	Live	client sidebar/components
Client communication workspace	Use email/WhatsApp/chat in client context	Select client → compose/query/conversation	CRM access	client + message models	assignment/WhatsApp routes	CMS integration	Live	CRM pages
Extended panels/conversion	Extra My Clients areas and project conversion	Menu/panel exists but no complete action	Unknown	none	none traced	TMS linkage absent	Stub/Planned	"coming soon" panels; no CRM → task service

3.13 Task Management System (TMS)

Module purpose

Provides admin workspaces/spaces and employee tasks, with a newer task dashboard built directly on communication spaces.

Users and roles

Admins manage workspaces; employees/space members create and complete work; creators review completion.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Admin workspace/space	CRUD workspaces/spaces, members and visibility	Admin configures workspace	Admin	Workspace, TaskSpace	/api/admin/workspace, /admin/space	task portal	Live	TaskManagement pages/controller
Legacy employee tasks/content	CRUD/assign tasks plus folders/docs/bookmarks	Employee opens workspace → manages content	workspace membership	Task, task content models	/api/employee/task/*	duplicated portal contexts	Partially implemented	contexts and employeeTask routes
Chat-space tasks	CRUD tasks in shared chat space	Space member opens TMS → task list	chat membership	ChatTask, Space	/api/chat/spaces/:id/tasks, /chat/tasks/:id	CMS is data source	Live	task dashboard WorkspaceContext.tsx
Assignment/review/comments	Multi-assign, due date, completion review and comments	Assignee completes → creator review	creator/assignee/member	ChatTask, ChatTaskComment	chat task/comment endpoints	sockets	Live	chat controller/models/components
Views/activity	List works; other views and task-specific audit incomplete	Select view/activity	member	tasks/assignment activity	UI + assignment activity API	none	Partially implemented	ClickupSpace.tsx, ActivityView.tsx
Task AI/content TODOs	Generate description, templates/tags/advanced options	Buttons/options not wired	member	none	no backend	future AI	Planned	TODOs in CreateTaskDialog.tsx copies

3.14 Requests and approvals

Module purpose

Aggregates heterogeneous employee requests and routes human decisions.

Users and roles

Employees submit/track; managers/HR/admin approve according to request type; admins' own requests may auto-approve.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Employee open requests	Aggregate own items and approvals	Open panel → filter → act	own/supervisory	many request models	/api/open-requests/*, unified request API	badges/sockets	Live	OpenRequestsPanel copies
Admin request centre	Review all supported request categories	Request Center → category → decide	admin/HR	request models	type-specific update endpoints	notifications	Live	admin RequestCenter/AllRequests
Generic edit/withdraw	Edit/withdraw using type mapping	User chooses action	own pending request	many models	generic open-request endpoints	none	Partially implemented	mapping cannot guarantee every type parity
Admin auto-approval	Resolve an admin's own submission	Submit as admin → approved fields applied	admin/owner	request records	request controller utility	automatic	Live	requestAutoApproval.js
Whistle reporting	Submit/review confidential report	Employee submits → admin reviews	employee/admin	WhistleReport	whistle routes	request centre	Partially implemented	anonymity/security need live verification

3.15 Shared platform administration

Module purpose

Provides tenant settings, company identity, users, notifications, search, files and import/export foundations.

Users and roles

Admin/super-admin configure; employees consume shared capabilities.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Company/users/settings	Configure company, admin accounts, time/date/font/features	Admin settings → save	admin/super-admin; personal font	CompanyProfile, User, settings models	company/user/settings routes	used by documents/payroll/UI	Partially implemented	many admin settings pages; time zone also hard-coded
Notifications	Real-time domain events and unread counts	Event occurs → socket/widget/badge updates	tenant/employee/room scopes	domain records, no canonical notification entity	Socket.IO and unread endpoints	realtime	Live	several io.on(connection) blocks and widgets
Search	Search within clients/messages/employees/shared content	User enters term in domain UI	domain access	domain models	multiple /search endpoints	none	Partially implemented	no global search surface/index
File management	Upload/download attachments and documents	Attach → stored path → authorised retrieval	domain permissions	domain attachment fields	Multer and /uploads	PDF/image processing	Live	upload middleware and routes
Bulk/general import-export	Bulk docs/offers work; universal tabular exchange is unclear	Admin uploads/downloads	admin	multiple	bulk endpoints	XLSX/CSV MIME support	Partially implemented	no general employee import/export flow traced
Audit/compliance	Domain histories exist but no unified immutable audit	Admin visits domain log	domain-specific	attendance/profile/message histories	domain routes	none	Unclear	no canonical audit log model

3.16 Reporting and analytics

Module purpose

Provides operational dashboards and domain summaries for workforce, time, leave and payroll.

Users and roles

Admin/HR/payroll users see organisation data; managers and employees see scoped/personal summaries.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
Workforce dashboard	Display headcount/milestones/events/request state	Admin opens Dashboard	tenant scope	employee/event/request	multiple APIs	widgets	Live	admin Dashboard/MainPage
Attendance analytics	Personal/admin stats and logs	Filter employee/period	own or attendance scope	attendance/logs	attendance stats APIs	payroll	Live	attendance pages/utilities
Leave analytics	Balance, usage and history	Open leave/profile/admin summary	own/HR	leave ledgers	summary/history APIs	payroll	Live	leave tables/pages
Payroll analytics	Estimates, period totals and slip summaries	Select period → calculate/review	payroll scope	estimate/slip/period	payroll APIs	finance inputs	Live	PayrollEstimate and PayrollPeriod pages
Exports	Selected PDFs/XLSX/CSV but no unified report catalogue	Use available domain export	domain scope	domain data	PDF/XLSX utilities	document generation	Partially implemented	employee PDF placeholder and uneven exports

3.17 AI and automation

Module purpose

Automates extraction/classification and scheduled business processing. It is not yet a configurable cross-module workflow product.

Users and roles

Mostly system jobs; HR benefits from extracted identity and processed email; admins configure source business settings rather than automation rules.

Function	What it does	User journey	Roles/permissions	Data entities	API/service	Automation/integration	Status	Code evidence
CNIC extraction	Sends image to vision model and parses fixed JSON	Mail/upload contains image → fields extracted	system/HR	onboarding employee data	extractCNICUsingOpenAI	OpenAI vision	Partially implemented	deepseekService.js, watcher.js
Email classification	Maps mail to five HR intents	Incoming mail → classifier → response branch	system	mail/offer/leave context	classifyEmailUsingOpenAI	OpenAI chat	Partially implemented	service and watcher
Draft/auto replies	Produce/respond to selected inbound mail	Intent branch → generated/templated response	system	mail records	draft/mail services	SMTP/OpenAI	Prototype	draftReply.js, watcher branches
Policy interpretation	Parse a few textual policy patterns	HR policy → regex-derived fields	HR/system	HR policy	policyParser.js	natural tokenizer	Prototype	fixed patterns/default 22
Scheduled business actions	Hard-coded attendance/leave/probation/message jobs	Clock trigger → tenant loop → mutation	system	domain models	node-cron	SMTP/sockets	Live	schedulers and four inline crons
Configurable workflow builder	User-defined triggers/conditions/actions	No implemented journey	none	none	none	none	Disabled	no workflow definition/executor/editor found